Standard Model quantum numbers from the $E_6 \times SU(3)$ chirality pair

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Abstract

Starting only from the 240 roots of E_8, we recover the branching $27 \to 1 + 16 + 10$ under $SO(10) \times U(1)$, the Standard Model content of the 16, the hypercharge generator, and the vanishing of the linear and cubic $U(1)$ sums. Every statement is an exact rational computation; see the scripts in `lisi/`. Nothing here is a derivation of the physical Standard Model, only a check that the chirality-theorem pair contains it.

1 Setup

Roots of E_8 are stored in doubled integer coordinates. The horizontal $SU(3)_M$ is the A_2 spanned by two roots with inner product -1, and E_6 is its centralizer. The 240 roots split as $72 + 6 + 6 \cdot 27$, the root form of

$$248 = (78, 1) + (1, 8) + (27, 3) + (\overline{27}, \bar{3}).$$

2 From E_6 to $SO(10)$

Fix one weight w_0 of the 27 (in true coordinates, $\langle w_0, w_0 \rangle = \frac{4}{3}$):

$$w_0 = \left[-\tfrac{2}{3} \quad -\tfrac{2}{3} \quad -\tfrac{2}{3} \quad 0 \quad 0 \quad 0 \quad 0 \quad 0\right].$$

The E_6 roots orthogonal to w_0 form a system of 40 roots, that of $SO(10)$, and the 27 splits by $\langle w, w_0 \rangle$:

$$27 = \underbrace{\mathbf{1}}_{4/3} + \underbrace{\mathbf{16}}_{1/3} + \underbrace{\mathbf{10}}_{-2/3}.$$

3 Standard Model content

Inside $SO(10)$ we select an $A_2 \times A_1$ subsystem (roots a, b and g) for which the 16 branches as one Standard Model generation:

$$a = \left[0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0\right], \quad b = \left[0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 1 \quad 0 \quad 0\right], \quad g = \left[0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 1\right].$$

The hypercharge generator is the vector orthogonal to $A_2 \times A_1$, to w_0 and to the horizontal A_2 that reproduces the Standard Model values; a weight w has $Y = \langle w, Y \rangle$ and $T_3 = \langle w, g \rangle / 2$ with

$$Y = \left[0 \quad 0 \quad 0 \quad \tfrac{1}{3} \quad -\tfrac{1}{3} \quad \tfrac{1}{3} \quad \tfrac{1}{2} \quad -\tfrac{1}{2}\right].$$

The 16

field	$\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L$	Y	$Q = T_3 + Y$	weights
Q_L	$(\mathbf{3}, \mathbf{2})$	$\frac{1}{6}$	$-\frac{1}{3},\ \frac{2}{3}$	6
u^c	$(\overline{\mathbf{3}}, \mathbf{1})$	$-\frac{2}{3}$	$-\frac{2}{3}$	3
d^c	$(\overline{\mathbf{3}}, \mathbf{1})$	$\frac{1}{3}$	$\frac{1}{3}$	3
L	$(\mathbf{1}, \mathbf{2})$	$-\frac{1}{2}$	$-1,\ 0$	2
ν^c	$(\mathbf{1}, \mathbf{1})$	0	0	1
e^c	$(\mathbf{1}, \mathbf{1})$	1	1	1

The 10 and the singlet

field	$\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L$	Y	$Q = T_3 + Y$	weights
D	$(\mathbf{3}, \mathbf{1})$	$-\frac{1}{3}$	$-\frac{1}{3}$	3
D^c	$(\overline{\mathbf{3}}, \mathbf{1})$	$\frac{1}{3}$	$\frac{1}{3}$	3
H_d	$(\mathbf{1}, \mathbf{2})$	$-\frac{1}{2}$	$-1,\ 0$	2
H_u	$(\mathbf{1}, \mathbf{2})$	$\frac{1}{2}$	$0,\ 1$	2

The singlet has $Y = 0$ and $Q = 0$.

Cross-check. On the 40 roots of $SO(10)$ this Y takes the values of the $SU(5)$ adjoint pattern: eight roots at 0, twelve at $\pm 5/6$, twelve at $\pm 1/6$, six at $\pm 2/3$ and two at ± 1. This is the standard hypercharge, not a flipped one.

4 Anomaly sums

Exact sums over the **16**:

$$\sum Y = 3\left(-\frac{2}{3}\right) + 2\left(-\frac{1}{2}\right) + 1 \cdot 0 + 6 \cdot \frac{1}{6} + 3 \cdot \frac{1}{3} + 1 \cdot 1 = 0,$$

$$\sum Y^3 = 3\left(-\frac{8}{27}\right) + 2\left(-\frac{1}{8}\right) + 1 \cdot 0 + 6 \cdot \frac{1}{216} + 3 \cdot \frac{1}{27} + 1 \cdot 1 = 0.$$

The **10** is vector-like, so its sums vanish termwise; the **1** carries $Y = 0$. The sums $\sum Q$ and $\sum Q^3$ over the 16 also vanish, as the scripts verify. This is the Standard Model gauge-anomaly cancellation within a single generation, visible here as a property of the E_6 weights.

5 Three families and triality

In the chirality theorem the representation is $V = 27 \boxtimes 3$, so the net chiral content is three copies of 16, i.e. 48 Weyl fermions, together with three vector-like 10's and three singlets. A separate check places the second D_4 of $D_4 \times D_4 \subset E_8$ inside E_6, giving $27 = 8_v + 8_s + 8_c + 3 \cdot \mathbf{1}$, the triality triple used by Lisi for generations. Whether the horizontal 3 and the triality 3 are related is an open question that these computations do not settle.

6 Reproducing

```
python3 lisi/run_all.py          # all checks
python3 lisi/make_report.py      # regenerate this PDF
```